

Address Normalization

Parsing, reference matching and review-ready output

Python / pandas / SQLite / YAML / Excel

THE PROBLEM

Address records arrive with inconsistent abbreviations, incomplete components and conflicting regional information. They need a repeatable preparation process before matching and downstream use.

MY ROLE

Created the solution and integrated it into working processes, including processing logic, configurable rules and output preparation.

THE SOLUTION

Created and integrated an address-processing engine that parses region, locality, street and building components; applies configurable normalization rules; and matches records against reference data. SQLite lookups and bounded caches support reference access.

Raw addresses

Parse + normalize

Reference match

Excel + review flags

DELIVERABLE

An Excel output with normalized fields, matching methods and data-quality flags. Uncertain or conflicting records remain identifiable for review.

VERIFICATION

Six focused local checks passed on the reviewed copy: five regression tests plus a SQLite integration-boundary test. They cover missing reference data, regional conflicts, village/building parsing, Excel output, lookup ambiguity and read-only reference access.

SCOPE

Checks use synthetic inputs. They do not establish production-wide matching accuracy or throughput. Client datasets and internal configuration are not included.